

Concepts of Stochastic Convergence

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Convergence in Probability

Limit Theorems

In this section, we will discuss two important theorems in probability, *the law of large numbers (LLN)* and *the central limit theorem (CLT)*.

The LLN basically states that the average of a large number of i.i.d. random variables converges to the expected value.

The CLT states that, under some conditions, the sum of a large number of random variables has an approximately normal distribution.

Law of Large Numbers

- The **law of large numbers** has a very central role in probability and statistics.
- It states that if you repeat an experiment independently a large number of times and average the result, what you obtain should be close to the expected value.
- There are two main versions of the law of large numbers.
- They are called the **weak** and **strong** laws of the large numbers. The difference between them is mostly theoretical. In this section, we state and prove the weak law of large numbers (WLLN).

Definition For i.i.d. random variables $X_1, X_2, \dots, X_n$, the **sample mean**, denoted by $\bar{X}$, is defined as

$$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n}.$$

Another common notation for the sample mean is M_n . If the X_i 's have CDF $F_X(x)$, we might show the sample mean by $M_n(X)$ to indicate the distribution of the X_i 's.

Cont.

Note that since the X_i 's are random variables, the sample mean, $\bar{X} = M_n(X)$, is also a random variable. In particular, we have

$$\begin{aligned} E[\bar{X}] &= \frac{EX_1 + EX_2 + \dots + EX_n}{n} && \text{(by linearity of expectation)} \\ &= \frac{nEX}{n} && \text{(since } EX_i = EX) \\ &= EX. \end{aligned}$$

Cont.

Also, the variance of $\bar{X}$ is given by

$$\begin{aligned}\text{Var}(\bar{X}) &= \frac{\text{Var}(X_1 + X_2 + \dots + X_n)}{n^2} && \text{(since } \text{Var}(aX) = a^2 \text{Var}(X)\text{)} \\ &= \frac{\text{Var}(X_1) + \text{Var}(X_2) + \dots + \text{Var}(X_n)}{n^2} && \text{(since the } X_i\text{'s are independent)} \\ &= \frac{n \text{Var}(X)}{n^2} && \text{(since } \text{Var}(X_i) = \text{Var}(X)\text{)} \\ &= \frac{\text{Var}(X)}{n}.\end{aligned}$$

State and prove the weak law of large numbers (WLLN)

The weak law of large numbers (WLLN)

Let $X_1, X_2, \dots, X_n$ be i.i.d. random variables with a finite expected value $EX_i = \mu < \infty$.
Then, for any $\epsilon > 0$,

$$\lim_{n \rightarrow \infty} P(|\bar{X} - \mu| \geq \epsilon) = 0.$$

Cont.

Proof

The proof of the weak law of large number is easier if we assume $\text{Var}(X) = \sigma^2$ is finite. In this case we can use Chebyshev's inequality to write

$$\begin{aligned} P(|\bar{X} - \mu| \geq \epsilon) &\leq \frac{\text{Var}(\bar{X})}{\epsilon^2} \\ &= \frac{\text{Var}(X)}{n\epsilon^2}, \end{aligned}$$

which goes to zero as $n \rightarrow \infty$.

Central Limit Theorem

- The **central limit theorem (CLT)** is one of the most important results in probability theory.
- It states that, under certain conditions, the sum of a large number of random variables is approximately normal.
- Here, we state a version of the CLT that applies to i.i.d. random variables.

Suppose that $X_1, X_2, \dots, X_n$ are i.i.d. random variables with expected values $EX_i = \mu < \infty$ and variance $\text{Var}(X_i) = \sigma^2 < \infty$. Then as we saw above, the sample mean $\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n}$ has mean $E\bar{X} = \mu$ and variance $\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$. Thus, the normalized random variable

$$Z_n = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{X_1 + X_2 + \dots + X_n - n\mu}{\sqrt{n}\sigma}$$

has mean $EZ_n = 0$ and variance $\text{Var}(Z_n) = 1$. The central limit theorem states that the CDF of Z_n converges to the standard normal CDF.

The Central Limit Theorem (CLT)

Let $X_1, X_2, \dots, X_n$ be i.i.d. random variables with expected value $EX_i = \mu < \infty$ and variance $0 < \text{Var}(X_i) = \sigma^2 < \infty$. Then, the random variable

$$Z_n = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{X_1 + X_2 + \dots + X_n - n\mu}{\sqrt{n}\sigma}$$

converges in distribution to the standard normal random variable as n goes to infinity, that is

$$\lim_{n \rightarrow \infty} P(Z_n \leq x) = \Phi(x), \quad \text{for all } x \in \mathbb{R},$$

where $\Phi(x)$ is the standard normal CDF.

Cont.

An interesting thing about the CLT is that it does not matter what the distribution of the X_i 's is the X_i 's can be discrete, continuous, or mixed random variables.

Example 1

Let's assume that X_i 's are *Bernoulli*(p). Then $Ex_i = p$, $Var(x_i) = p(1 - p)$. Also , $Y_n = X_1 + X_2 + \dots + X_n$ has *Binomial* (n, p) distribution. Therefore,

$$Z_n = \frac{Y_n - np}{\sqrt{np(1 - p)}},$$

Assumptions:

- $X_1, X_2 \dots$ are iid $\text{Bernoulli}(p)$.
- $Z_n = \frac{X_1 + X_2 + \dots + X_n - np}{\sqrt{np(1-p)}}$.

We choose $p = \frac{1}{3}$.

$$Z_1 = \frac{X_1 - p}{\sqrt{p(1-p)}}$$

PMF of Z_1



$$Z_2 = \frac{X_1 + X_2 - 2p}{\sqrt{2p(1-p)}}$$

PMF of Z_2



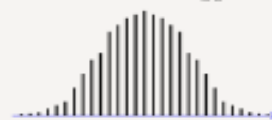
$$Z_3 = \frac{X_1 + X_2 + X_3 - 3p}{\sqrt{3p(1-p)}}$$

PMF of Z_3



$$Z_{30} = \frac{\sum_{i=1}^{30} X_i - 30p}{\sqrt{30p(1-p)}}$$

PMF of Z_{30}



Z_n is the normalized sum of n independent $\text{Bernoulli}(p)$ random variables. The shape of its PMF, $P_{Z_n}(z)$, resembles the normal curve as n increases.

Example 2

Let's assume that X_i 's are *Uniform*(0,1), Then $Ex_i = 1/2$, $Var(x_i) = 1/12$.

In this case,

$$Z_n = \frac{X_1 + X_2 + \dots + X_n - \frac{n}{2}}{\sqrt{n/12}}.$$

Assumptions:

- $X_1, X_2 \dots$ are iid $\text{Uniform}(0,1)$.
- $Z_n = \frac{X_1 + X_2 + \dots + X_n - \frac{n}{2}}{\sqrt{\frac{n}{12}}}$.

$$Z_1 = \frac{X_1 - \frac{1}{2}}{\sqrt{\frac{1}{12}}}$$

PDF of Z_1



$$Z_2 = \frac{X_1 + X_2 - 1}{\sqrt{\frac{2}{12}}}$$

PDF of Z_2



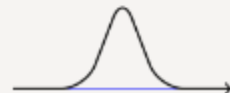
$$Z_3 = \frac{X_1 + X_2 + X_3 - \frac{3}{2}}{\sqrt{\frac{3}{12}}}$$

PDF of Z_3



$$Z_{30} = \frac{\sum_{i=1}^{30} X_i - \frac{30}{2}}{\sqrt{\frac{30}{12}}}$$

PDF of Z_{30}



Z_n is the normalized sum of n independent $\text{Uniform}(0,1)$ random variables. The shape of its PDF, $f_{Z_n}(z)$, gets closer to the normal curve as n increases.

Real Applications

- Laboratory measurement errors are usually modeled by normal random variables.
- In communication and signal processing, Gaussian noise is the most frequently used model for noise.
- In finance, the percentage changes in the prices of some assets are sometimes modeled by normal random variables.
- When we do random sampling from a population to obtain statistical knowledge about the population, we often model the resulting quantity as a normal random variable.

How to apply the Central Limit Theorem

Here are the steps that we need in order to apply the CLT:

1. Write the random variable of interest, Y , as the sum of n i.i.d. random variable X_i 's:

$$Y = X_1 + X_2 + \dots + X_n.$$

2. Find EY and $\text{Var}(Y)$ by noting that

$$EY = n\mu, \quad \text{Var}(Y) = n\sigma^2,$$

where $\mu = EX_i$ and $\sigma^2 = \text{Var}(X_i)$.

3. According to the CLT, conclude that $\frac{Y - EY}{\sqrt{\text{Var}(Y)}} = \frac{Y - n\mu}{\sqrt{n}\sigma}$ is approximately standard normal; thus, to find $P(y_1 \leq Y \leq y_2)$, we can write

$$\begin{aligned} P(y_1 \leq Y \leq y_2) &= P\left(\frac{y_1 - n\mu}{\sqrt{n}\sigma} \leq \frac{Y - n\mu}{\sqrt{n}\sigma} \leq \frac{y_2 - n\mu}{\sqrt{n}\sigma}\right) \\ &\approx \Phi\left(\frac{y_2 - n\mu}{\sqrt{n}\sigma}\right) - \Phi\left(\frac{y_1 - n\mu}{\sqrt{n}\sigma}\right). \end{aligned}$$

Example 3

A bank teller serves customers standing in the queue one by one. Suppose that the service time X_i for customer i has mean $EX_i = 2$ (minutes) and $\text{Var}(X_i) = 1$. We assume that service times for different bank customers are independent. Let Y be the total time the bank teller spends serving 50 customers. Find $P(90 < Y < 110)$.

Example 4

In a communication system each data packet consists of 1000 bits. Due to the noise, each bit may be received in error with probability 0.1. It is assumed bit errors occur independently. Find the probability that there are more than 120 errors in a certain data packet.

Continuity Correction

Let us assume that $Y \sim \text{Binomial}(n = 20, p = \frac{1}{2})$, and suppose that we are interested in $P(8 \leq Y \leq 10)$. We know that a $\text{Binomial}(n = 20, p = \frac{1}{2})$ can be written as the sum of n i.i.d. $\text{Bernoulli}(p)$ random variables:

$$Y = X_1 + X_2 + \dots + X_n.$$

Since $X_i \sim \text{Bernoulli}(p = \frac{1}{2})$, we have

$$EX_i = \mu = p = \frac{1}{2}, \quad \text{Var}(X_i) = \sigma^2 = p(1 - p) = \frac{1}{4}.$$

Cont.

Thus, we may want to apply the CLT to write

$$\begin{aligned}P(8 \leq Y \leq 10) &= P\left(\frac{8 - n\mu}{\sqrt{n}\sigma} < \frac{Y - n\mu}{\sqrt{n}\sigma} < \frac{10 - n\mu}{\sqrt{n}\sigma}\right) \\&= P\left(\frac{8 - 10}{\sqrt{5}} < \frac{Y - n\mu}{\sqrt{n}\sigma} < \frac{10 - 10}{\sqrt{5}}\right) \\&\approx \Phi(0) - \Phi\left(\frac{-2}{\sqrt{5}}\right) \\&= 0.3145\end{aligned}$$

Cont.

Since, here, $n = 20$ is relatively small, we can actually find $P(8 \leq Y \leq 10)$ accurately. We have

$$\begin{aligned} P(8 \leq Y \leq 10) &= \sum_{k=8}^{10} \binom{n}{k} p^k (1-p)^{n-k} \\ &= \left[\binom{20}{8} + \binom{20}{9} + \binom{20}{10} \right] \left(\frac{1}{2} \right)^{20} \\ &= 0.4565 \end{aligned}$$

We notice that our approximation is not so good. Part of the error is due to the fact that Y is a discrete random variable and we are using a continuous distribution to find $P(8 \leq Y \leq 10)$. Here is a trick to get a better approximation, called **continuity correction**. Since Y can only take integer values, we can write

$$\begin{aligned}P(8 \leq Y \leq 10) &= P(7.5 < Y < 10.5) \\&= P\left(\frac{7.5 - n\mu}{\sqrt{n}\sigma} < \frac{Y - n\mu}{\sqrt{n}\sigma} < \frac{10.5 - n\mu}{\sqrt{n}\sigma}\right) \\&= P\left(\frac{7.5 - 10}{\sqrt{5}} < \frac{Y - n\mu}{\sqrt{n}\sigma} < \frac{10.5 - 10}{\sqrt{5}}\right) \\&\approx \Phi\left(\frac{0.5}{\sqrt{5}}\right) - \Phi\left(\frac{-2.5}{\sqrt{5}}\right) \\&= 0.4567\end{aligned}$$

As we see, using continuity correction, our approximation improved significantly. The continuity correction is particularly useful when we would like to find $P(y_1 \leq Y \leq y_2)$, where Y is binomial and y_1 and y_2 are close to each other.

Continuity Correction for Discrete Random Variables

Let $X_1, X_2, \dots, X_n$ be independent discrete random variables and let

$$Y = X_1 + X_2 + \dots + X_n.$$

Suppose that we are interested in finding $P(A) = P(l \leq Y \leq u)$ using the CLT, where l and u are integers. Since Y is an integer-valued random variable, we can write

$$P(A) = P(l - \frac{1}{2} \leq Y \leq u + \frac{1}{2}).$$

It turns out that the above expression sometimes provides a better approximation for $P(A)$ when applying the CLT. This is called the continuity correction and it is particularly useful when X_i 's are Bernoulli (i.e., Y is binomial).

Problem 1

There are 100 men on a plane. Let X_i be the weight (in pounds) of the i th man on the plane. Suppose that the X_i 's are i.i.d., and $EX_i = \mu = 170$ and $\sigma_{X_i} = \sigma = 30$. Find the probability that the total weight of the men on the plane exceeds 18,000 pounds.

Problem 2

Let $X_1, X_2, \dots, X_{25}$ be i.i.d. with the following PMF

$$P_X(k) = \begin{cases} 0.6 & k = 1 \\ 0.4 & k = -1 \\ 0 & \text{otherwise} \end{cases}$$

And let

$$Y = X_1 + X_2 + \dots + X_n.$$

Using the CLT and continuity correction, estimate $P(4 \leq Y \leq 6)$.

Problem 3

You have invited 64 guests to a party. You need to make sandwiches for the guests. You believe that a guest might need 0, 1 or 2 sandwiches with probabilities $\frac{1}{4}$, $\frac{1}{2}$, and $\frac{1}{4}$ respectively. You assume that the number of sandwiches each guest needs is independent from other guests. How many sandwiches should you make so that you are 95% sure that there is no shortage?

Problem 4

Let $X_1, X_2, \dots, X_n$ be i.i.d. *Exponential*(λ) random variables with $\lambda = 1$. Let

$$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n}.$$

How large should n be such that

$$P\left(0.9 \leq \bar{X} \leq 1.1\right) \geq 0.95?$$